PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 107		CPT	

1st SEMESTER ***2024/ 25 RESIT*** EXAMINATION

DISCRETE MATHEMATICS AND STATISTICS

TIME ALLOWED: ***2 Hours***

INSTRUCTIONS TO CANDIDATES

1. This is a ***closed-book*** examination, ***which is to be written without books or notes***.
2. Total marks available are 100, accounting for 100% of the overall module marks.
3. The number in the column on the right indicates the marks for each question.
4. Answer all questions. There is NO penalty for providing a wrong answer.
5. Only English solutions are accepted. Answer should be written in the answer booklet(s) provided.
6. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Notes:

- **To obtain full marks for each question, relevant and clear steps need to be included in the answers.**
- **Partial marks may be awarded depending on the degree of completeness and clarity.**

Question 1: Proof Techniques **[12 marks]**

(a) Use proof by *contradiction* to show that there does not exist any integers a and b such that $3a + 18b = 1$.

(2 marks)

(b) $\sqrt{180}$ is irrational. If you think that it is true, prove it. If not, explain why.

(5 marks)

(c) Let $x \in \mathbb{Z}$ and $1 \leq x$, use proof by induction to show that:

$$((2x)^2 + \ldots + 3^2 + 2^2 + 1) = (\tfrac{1}{3})(1 + 2x)(4x + 1)x.$$

(5 marks)

Question 2: Set Theory **[18 marks]**

(a) Let the universal set be $\mathbb{N}$, $P = \{x \in \mathbb{N} \mid x \text{ is even and } 3 < x < 9\}$, $Q = \{x \in \mathbb{N} \mid x \text{ is odd and } 4 < x < 8\}$ and $R = \{x \in \mathbb{N} \mid x \text{ is odd and } 3 < x < 9\}$. Find the elements of:

1. $(R \cap P) \cup (\sim(\sim P \cap \sim Q))$
2. $R \,\Delta\, (P - Q)$
3. $((\sim(\sim P \cap \sim Q)) \cup (R \cap P)) \cap (R \,\Delta\, (P - Q))$

(6 marks)

(b) Let the universal set $U = \{2,3,4,5,6,7,8\}$, $P = \{x \in \mathbb{Z} \mid x \text{ is even and } 3 < x < 7\}$, $Q = \{y \in \mathbb{Z} \mid y \text{ is even and } 4 < y < 8\}$ and $R = \{t \in \mathbb{Z} \mid t \text{ is odd and } 4 < t < 9\}$. Find the characteristic vector of:

1. $(Q \cap R) \cup P$
2. $P \cup (Q - R)$
3. $(P \,\Delta\, R) \cap (Q \cap {\sim}P)$

(6 marks)

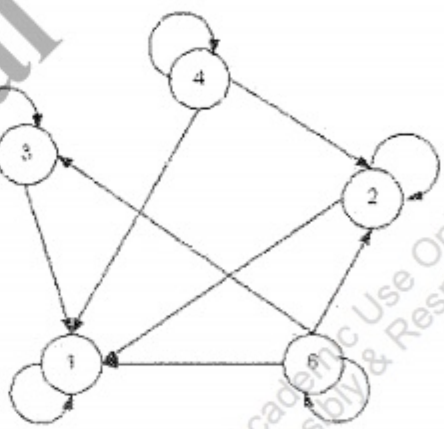
(c) If you think the following sets are equal, prove it. If not, explain why.

1. A x (B ∪ C) = (A x B) ∪ (A x C)
2. A x (B − C) = (A x B) − (A x C)

(6 marks)

Question 3: Relations **[10 marks]**

(a) Give the relation represented by the following digraph.



(5 marks)

(b) Let $M = \{d, c, b, a\}$ and $R \subseteq M \times M$ be given by $R = \{(c,d), (b,a), (a,a)\}$. Determine the inverse relation R^{-1} of R.

(5 marks)

Question 4: Functions **[12 marks]**

(a) Let $f: \mathbb{Z}^{+} \times \mathbb{Z} \rightarrow \mathbb{Q}$ be given by $f(x, y) = \frac{2x}{y}$. Prove or disprove that $f(x, y)$ is a bijective function.

(2 marks)

(b) $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = 5 - x^3$. Find the inverse of $f(x)$.

(6 marks)

(c) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = 3x^2 + 2$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be given by $g(x) = 4x - 5$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$ and $(g \circ g)(x)$.

(4 marks)

Question 5: Logic **[22 marks]**

(a) Let p, q and r stand for the following propositions:

- p: "You get a grade A."
- q: "You score at least 70% on the final exam."
- r: "The average of your coursework is above 90%."

Use p, q and r to translate the following into propositional logic:

1. You get a grade A only if you score at least 70% on the final exam or the average of your coursework is above 90%.
2. If you score at least 70% on the final exam or the average of your coursework is above 90%, then you will get a grade A.

(4 marks)

(b) Show whether the statements below are tautologies and/or contradictions:

1. $p \rightarrow (p \vee (q \vee r))$
2. $(q \wedge \neg p) \wedge (p \vee \neg q)$
3. $(p \rightarrow r) \rightarrow (p \wedge r)$

(6 marks)

(c) Let S be the signature consisting of the unary predicates *professor*, *module_leader*, *module*, the binary predicate symbol *teaches* and the individual constants *John* and *CPT107*. Assume that these symbols have the following meaning:

- *professor(x)* states: x is a professor.
- *module_leader(x)* states: x is a module leader.
- *teaches(x,y)* states: x teaches the module y.
- *module(x)* states: x is module.
- *John* and *CPT107* refer to "John" and "CPT107", respectively.

Translate the following sentences into first-order predicate logic:

1. CPT107 is a module.
2. Every professor is a module leader.
3. John teaches CPT107.
4. Every professor teaches a module.
5. John teaches a module.
6. Some professors are not module leaders.

(12 marks)

Question 6: Combinatorics and Probability **[26 marks]**

(a) There are 50 bags of oranges. Each bag consists of no more than 24 oranges. Show that there are at least 3 bags consisting of the same number of oranges.

(4 marks)

(b) A box consists of 4 green and 7 black balls. Balls are drawn uniformly at random. Each time, a ball is drawn from the box and the color of the ball is recorded. Then the ball will be put back into the box together with an extra ball of the same color. Answer the following questions:

1. What is the probability that the first and the second balls drawn are both red?
2. What is the probability that the first and the second balls drawn are both green?
3. What is the probability that at least one of the first two drawn balls drawn is green?
4. Given that exactly one of the first two drawn balls drawn is green, what is the probability that the third ball drawn is not green?

(12 marks)

(c) There are 9 boys and 5 girls.

1. How many ways are there for them to be seated in a round table without any restriction?
2. How many ways are there for them to be seated in a round table under the condition that 5 girls sit together?
3. How many ways are there for them be seated in a round table under the condition that no 2 girls sit next to each other?

(6 marks)

(d) Rena and Larson are playing games. Rena rolls a regular six-sided dice.
If the dice lands on an even number, Rena gets \$1 from Larson. However, if the dice lands on an odd number, Rena must give \$1 to Larson. Either player could win \$1 on each roll of the dice. What is the expected profit of Rena?

(4 marks)

END OF RESIT EXAM PAPER